

Physics I — Units, Kinematics, Motion, Energy and Fluids

Chapter F9.1 — A Recall Layer, Not a Teaching Layer, Because This Is Your Undergraduate Foundation Handed Back

Module	Foundation Layer — Module 9 — General Science & Computer Applications
Chapter	Chapter F9.1
Topic	Official syllabus head 6 — General Science
Version	v1.0
Compiled	16 August 2026
Exam stage	Recruitment Test (RT)

NOTE · WHY THIS CHAPTER IS SHAPED DIFFERENTLY FROM MODULE 6

Every previous Foundation chapter taught from zero, because accounting and auditing were genuinely new to you. **This one does not**, and that is deliberate.

Blueprint §9.1 is explicit: the General Science half of this module is **physics-dominated**, and for a Mechanical Engineer *"this is not new learning — it is recall."* Teaching you kinematics from first principles would burn hours the Blueprint exists to save.

So F9.1 and F9.2 are **recall layers**: dense tables, the formulas that get asked, the traps that catch people who half-remember, and heavy practice. Read the tables, then go straight to the questions. If a question exposes a gap, come back to that one table.

HIGH YIELD · WHY MODULE 9 EXISTS AT ALL

This module was **not in your original plan**. Blueprint §9.1 recommended adding it, and called it *"the single correction worth more marks than any other change available to you."*

The evidence: **General Science and Computer Applications is official syllabus head 6**, it appeared in **every** paper analysed, and it has **never been worth less than about 28 marks**. In 2023 it was worth **50 marks — as much as the entire labour law block** (PYQ-DERIVED, Blueprint §9.1).

And the 2023 science questions were: **combined lens power, an inelastic collision, the uniform acceleration of a train, a concave mirror with the object at infinity**, plus effluents and pH, soaps, eyeball layers, an enthalpy reaction, banyan prop roots and ocean acidification.

Four of those ten are mechanics and optics you have solved hundreds of times. That is the argument for this module in one line.

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0. How This Chapter Works

STUDY SESSION · CHAPTER F9.1 — FIVE SHORT SESSIONS

1. **Session 1** — Units, measurement, scalars and vectors (§1) ★★★★★
2. **Session 2** — Kinematics and the equations of motion (§2) ★★★★★
3. **Session 3** — Laws of motion, momentum and **collisions** (§3) ★★★★★
4. **Session 4** — Work, energy and power (§4) ★★★★★
5. **Session 5** — Gravitation and fluids (§5) ★★★★★

Then 30 graded questions in §6. Because this is recall, budget **two days**, not four — and spend the larger share of it on the questions rather than the text.

1. Session 1 — Units, Scalars and Vectors ★★★★★

1.1 The seven SI base units

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Seven base units — the count itself is asked

1.2 Derived units worth knowing by name

Unit	Quantity	In base units
newton (N)	Force	kg m s^{-2}
joule (J)	Work, energy	N m
watt (W)	Power	J s^{-1}
pascal (Pa)	Pressure	N m^{-2}
hertz (Hz)	Frequency	s^{-1}
coulomb (C)	Electric charge	A s
diopetre (D)	Power of a lens	m^{-1}

1.3 Scalars and vectors

Scalars have magnitude only: mass, **speed**, distance, **work**, **energy**, power, time, temperature, density, pressure, electric charge.

Vectors have magnitude and direction: displacement, **velocity**, acceleration, **force**, **momentum**, **weight**, impulse, torque.

MUST MASTER · THE THREE PAIRS THE EXAM USES

1. **Work and energy are SCALARS**, even though work is defined using force, which is a vector. The dot product of two vectors is a scalar. This is the single most common wrong option in the topic
2. **Speed is scalar, velocity is vector**. Likewise **distance** scalar, **displacement** vector
3. **Mass is a scalar and is constant. Weight is a vector and varies** with location, because $W = mg$ and g varies

Pressure is also a scalar, which surprises people, because although force has direction, pressure at a point acts equally in all directions.

CHECK YOURSELF · SESSION 1

1. Name the seven SI base units and their quantities
2. Express the newton, joule, watt and pascal in base units
3. Are work and energy scalars or vectors? Why?
4. Which of mass and weight is constant, and why does the other vary?

2. Session 2 — Kinematics ★★★★★

2.1 The equations of motion

For **uniform (constant) acceleration**:

Equation	Does not contain
$v = u + at$	displacement s
$s = ut + \frac{1}{2}at^2$	final velocity v
$v^2 = u^2 + 2as$	time t
$s = [(u + v)/2] \times t$	acceleration a

Choose the equation by what the question does not give you

Where u is initial velocity, v final velocity, a acceleration, t time and s displacement.

For **free fall**, $a = g$ about 9.8 m s^{-2} downwards (take 10 where the question permits). Thrown upwards, use $a = -g$.

2.2 Graphs

Graph	Slope gives	Area under gives
Distance–time	Speed	—
Velocity–time	Acceleration	Displacement
Acceleration–time	—	Change in velocity

MUST MASTER · THE UNIT CONVERSION THAT DECIDES NUMERICALS

To convert km/h into m/s, multiply by 5/18. To go the other way, multiply by 18/5.

So **72 km/h = 20 m/s**, and **36 km/h = 10 m/s**. Nearly every kinematics numerical in this examination is set in km/h and must be solved in SI, and the 2023 paper's question on the **uniform acceleration of a train** is exactly this shape.

Do the conversion first, before you choose an equation. The commonest error is not the physics — it is solving correctly in the wrong units.

Worked, in the 2023 style: a train starting from rest reaches 72 km/h in 20 s under uniform acceleration.
 $72 \text{ km/h} = 72 \times 5/18 = 20 \text{ m s}^{-1}$. Then $a = (v - u)/t = (20 - 0)/20 = \mathbf{1 \text{ m s}^{-2}}$

CHECK YOURSELF · SESSION 2

1. Write the four equations of motion and say which quantity each omits
2. Convert 90 km/h to m/s, and 25 m/s to km/h
3. What does the slope of a velocity–time graph give? What does the area give?
4. A body is dropped from rest. How far does it fall in 3 s, taking $g = 10 \text{ m s}^{-2}$?

3. Session 3 — Laws of Motion and Collisions ★★★★★

3.1 Newton's three laws

Law	Statement	What it gives us
First	A body continues at rest or in uniform motion unless acted on by an external force	Defines inertia , and defines force qualitatively
Second	Force equals the rate of change of momentum; $F = ma$ when mass is constant	Defines force quantitatively , and gives the newton
Third	To every action there is an equal and opposite reaction	Pairs of forces, and conservation of momentum

MUST MASTER · TWO POINTS ABOUT THE THIRD AND SECOND LAWS

- Action and reaction act on DIFFERENT bodies**, and therefore **never cancel each other**. A statement that they cancel, or that they act on the same body, is wrong. This is the single most examinable subtlety in the laws of motion
- The general form of the second law is $F = dp/dt$** , force equals the rate of change of momentum. **$F = ma$ is the special case where mass is constant**. So an option saying the second law applies only when mass is constant is wrong — it is $F = ma$ that carries that restriction, not the law

Three kinds of inertia are sometimes asked by name: inertia of **rest**, of **motion**, and of **direction**.

Momentum $p = mv$, a vector, unit kg m s^{-1} . **Impulse** $= F \times t = \text{change in momentum}$, unit N s .

3.2 Collisions — the 2023 pattern

	Elastic	Inelastic	Perfectly inelastic
Momentum conserved	Yes	Yes	Yes
Kinetic energy conserved	Yes	No	No — loss is maximum
After impact	Bodies separate	Bodies separate	Bodies move together
Coefficient of restitution e	$e = 1$	$0 < e < 1$	$e = 0$

MUST MASTER · MOMENTUM IS CONSERVED IN EVERY COLLISION

Linear momentum is conserved in all collisions. Kinetic energy is conserved only in an elastic collision.

That one sentence answers most collision questions, and the 2023 paper's inelastic-collision item is built on it. The trap is an option asserting that momentum is not conserved in an inelastic collision — it always is, because conservation of momentum follows from Newton's third law and holds regardless of whether energy is lost to heat, sound or deformation.

Coefficient of restitution $e = (\text{relative velocity of separation}) / (\text{relative velocity of approach})$.
 $e = 1$ perfectly elastic, $e = 0$ perfectly inelastic.

Worked, perfectly inelastic: 2 kg at 5 m s^{-1} strikes a stationary 3 kg body and they move together. Momentum before = $2 \times 5 = 10 \text{ kg m s}^{-1}$. After, $(2 + 3)v = 10$, so **$v = 2 \text{ m s}^{-1}$** . Check the energy: before = $\frac{1}{2} \times 2 \times 25 = 25 \text{ J}$; after = $\frac{1}{2} \times 5 \times 4 = 10 \text{ J}$. **15 J lost** — which is exactly why it is inelastic.

3.3 Circular motion

Centripetal force = mv^2/r , directed **towards** the centre — it is the real force that keeps a body on a circular path.

Centrifugal force is a **pseudo (fictitious) force**, apparent only in a rotating frame, directed **away** from the centre. Calling it a real force is a standard wrong option.

CHECK YOURSELF · SESSION 3

1. State the three laws. Which defines force quantitatively?
2. Do action and reaction cancel? Why not?
3. Which is the general form of the second law, and what does $F = ma$ assume?
4. Which quantity is conserved in all collisions? Which only in elastic ones?
5. Give e for perfectly elastic and perfectly inelastic collisions
6. Is centrifugal force real? Which way does it point?

4. Session 4 — Work, Energy and Power ★★★★★

4.1 Work

$W = F s \cos \theta$, where θ is the angle between force and displacement. Unit: **joule**. A **scalar**.

θ	$\cos \theta$	Work
0°	1	Maximum, positive
90°	0	Zero
180°	-1	Negative

MUST MASTER · WHEN WORK IS ZERO

Work is **zero when the force is perpendicular to the displacement**, because $\cos 90^\circ = 0$.

Three standard illustrations, each of which has appeared in one form or another:

- **Carrying a bucket while walking on level ground** — the force is vertical, the displacement

horizontal, so the work done against gravity is **zero**

- A **satellite in a circular orbit** — the gravitational force is towards the centre, the displacement

tangential, so gravity does **no work**

- A **coolie carrying a load on a level platform** — the same situation as the bucket

Note the counter-intuitive framing: the man is certainly getting tired, but the *physics* definition of work is zero. The exam relies on you confusing effort with work.

4.2 Energy and power

Quantity	Formula	Unit
Kinetic energy	$\frac{1}{2}mv^2$	joule
Potential energy	mgh	joule
Power	W/t	watt

- **Work–energy theorem:** the work done by the net force equals the **change in kinetic energy**
- **1 horsepower about 746 W**
- **The commercial unit of electrical energy is the kilowatt-hour: 1 kWh = 3,600,000 J**, that is

3.6 million joules

CHECK YOURSELF · SESSION 4

1. Give the work formula. When is work zero, and give two examples
2. Is work a scalar or a vector?
3. State the work–energy theorem
4. How many joules in a kilowatt-hour? How many watts in a horsepower?

5. Session 5 — Gravitation and Fluids ★★★★★

5.1 Gravitation

$$F = G m_1 m_2 / r^2 \text{ and } g = GM/R^2$$

Fact	Detail
g varies	Decreases with altitude and with depth ; zero at the centre of the Earth
g is maximum at the poles , minimum at the equator	Because the Earth bulges at the equator and rotates
Escape velocity from Earth	About 11.2 km s⁻¹
Orbital velocity near the surface	About 7.9 km s⁻¹
Relationship	Escape velocity = sqrt2 × orbital velocity

MUST MASTER · WEIGHTLESSNESS IS NOT THE ABSENCE OF GRAVITY

An astronaut in an orbiting satellite feels weightless because the astronaut **and** the satellite are both in **free fall** — they accelerate towards the Earth at the same rate, so there is no contact force between them.

Gravity is very much present; indeed it is what keeps the satellite in orbit. The option "there is no gravity in space" is wrong, and it is the intuitive answer.

The state is properly called **apparent weightlessness**, and the same thing happens in a lift whose cable is cut.

5.2 Fluids — four principles

Principle	Statement	Application
Pascal's law	Pressure applied to an enclosed fluid is transmitted equally in all directions	Hydraulic press, hydraulic brakes , hydraulic lift
Archimedes' principle	The upthrust equals the weight of the fluid displaced	Flotation, ships, submarines, hydrometer
Bernoulli's principle	Where the velocity of a fluid is high , its pressure is low	Aerofoil lift , atomiser, spin of a ball, Venturi meter
Law of flotation	A floating body displaces its own weight of fluid	Ships, icebergs

Pressure = Force / Area, unit **pascal**. Atmospheric pressure about 1.013×10^5 Pa, measured by a barometer.

COMMON UPSC TRAP · WHICH PRINCIPLE EXPLAINS WHICH DEVICE

The three fluid principles are constantly swapped in matching questions. Fix them by the device:

- **Hydraulic brakes and presses » Pascal** — transmission of pressure
- **Ships, submarines and hydrometers » Archimedes** — upthrust
- **Aeroplane lift, atomiser, swinging cricket ball » Bernoulli** — velocity and pressure

If the device *multiplies force*, it is Pascal. If it *floats or sinks*, it is Archimedes. If something *moves through or past a fluid*, it is Bernoulli.

CHECK YOURSELF · SESSION 5

1. Where is g maximum, and why?
2. Give escape and orbital velocities, and the relationship between them
3. Why does an astronaut feel weightless? What is the wrong explanation?
4. Match: hydraulic brake, ship, aerofoil — to their principles
5. State the law of flotation

6. Graded Practice — 30 Questions ★★★★★

6.1 Level 1 — Recognition

- Q1.** The number of base units in the SI system is (a) five (b) seven (c) nine (d) three
- Q2.** Which one of the following is a **vector** quantity? (a) Work (b) Energy (c) Speed (d) Momentum
- Q3.** The SI unit of force is the (a) newton (b) joule (c) watt (d) pascal
- Q4.** Newton's first law of motion defines (a) momentum (b) power (c) inertia (d) impulse
- Q5.** In **any** collision, the quantity that is always conserved is (a) kinetic energy (b) linear momentum (c) velocity (d) mechanical energy
- Q6.** The SI unit of pressure is the (a) pascal (b) newton (c) joule (d) watt
- Q7.** The work done is zero when the angle between the force and the displacement is (a) 0° (b) 45° (c) 180° (d) 90°
- Q8.** Kinetic energy is given by (a) mgh (b) mv (c) $\frac{1}{2}mv^2$ (d) Fs
- Q9.** The escape velocity from the surface of the Earth is about (a) 7.9 km s^{-1} (b) 11.2 km s^{-1} (c) 3.0 km s^{-1} (d) 30 km s^{-1}
- Q10.** The upthrust on a body immersed in a fluid equals the weight of the fluid displaced. This is (a) Archimedes' principle (b) Pascal's law (c) Bernoulli's principle (d) Hooke's law

6.2 Level 2 — Understanding

- Q11.** Which equation of motion does **not** involve time? (a) $v = u + at$ (b) $s = ut + \frac{1}{2}at^2$ (c) $v^2 = u^2 + 2as$ (d) $s = [(u + v)/2]t$
- Q12.** In a perfectly inelastic collision (a) kinetic energy is conserved (b) momentum is not conserved (c) the bodies separate with equal speeds (d) the bodies move together with a common velocity and the loss of kinetic energy is maximum
- Q13.** The coefficient of restitution for a perfectly elastic collision is (a) 1 (b) 0 (c) between 0 and 1 (d) infinite
- Q14.** Action and reaction (a) act on the same body and cancel each other (b) act on different bodies and therefore do not cancel (c) are unequal in magnitude (d) act in the same direction
- Q15.** The area under a velocity–time graph gives (a) acceleration (b) speed (c) displacement (d) force
- Q16.** An astronaut in an orbiting satellite feels weightless because (a) the astronaut and the satellite are both in free fall (b) there is no gravity in space (c) the astronaut's mass becomes zero (d) the satellite is beyond the atmosphere
- Q17.** The value of the acceleration due to gravity is (a) the same everywhere on the Earth's surface (b) maximum at the equator (c) independent of altitude (d) maximum at the poles and minimum at the equator

Q18. One kilowatt-hour is equal to (a) 3,600 J (b) 3,600,000 J (c) 1,000 J (d) 746 J

Q19. Pressure applied to an enclosed fluid is transmitted equally in all directions. This is (a) Archimedes' principle (b) Bernoulli's principle (c) Pascal's law (d) Boyle's law

Q20. The lift on an aeroplane wing is explained by (a) Bernoulli's principle (b) Pascal's law (c) Archimedes' principle (d) Newton's first law

6.3 Level 3 — Recruitment Test standard

Q21. A train starting from rest attains a speed of 72 km/h in 20 seconds with uniform acceleration. Its acceleration is (a) 3.6 m s^{-2} (b) 20 m s^{-2} (c) 0.5 m s^{-2} (d) 1 m s^{-2}

Q22. A body of mass 2 kg moving at 5 m s^{-1} collides with a stationary body of mass 3 kg, and the two move together after impact. Their common velocity is (a) 5 m s^{-1} (b) 2 m s^{-1} (c) 2.5 m s^{-1} (d) 1.5 m s^{-1}

Q23. Which one of the following statements is **not** correct? (a) Momentum is conserved in both elastic and inelastic collisions (b) Kinetic energy is conserved only in an elastic collision (c) Work and energy are vector quantities (d) Impulse is equal to the change in momentum

Q24. Consider the following statements:

1. Mass is a scalar and remains constant, whereas weight is a vector and varies with location.
2. The escape velocity from the Earth is about $\sqrt{2}$ times the orbital velocity near its surface.
3. Centrifugal force is a real force directed towards the centre.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

Q25. Match List-I with List-II and select the correct answer using the code given below:

List-I (Unit)	List-II (Quantity)
A. Newton	1. Pressure
B. Pascal	2. Force
C. Joule	3. Power
D. Watt	4. Energy

(a) A-1 B-2 C-3 D-4 (b) A-2 B-1 C-3 D-4 (c) A-1 B-2 C-4 D-3 (d) A-2 B-1 C-4 D-3

Q26. A body is thrown vertically upwards with a velocity of 20 m s^{-1} . Taking $g = 10 \text{ m s}^{-2}$, the maximum height it reaches is (a) 10 m (b) 20 m (c) 40 m (d) 5 m

Q27. The work done in carrying a bucket while walking horizontally along a level road is (a) equal to the weight multiplied by the distance (b) negative (c) zero (d) maximum

Q28. Consider the following statements about Newton's second law of motion:

1. It gives the SI unit of force.
2. In its general form it states that force equals the rate of change of momentum.
3. The law itself applies only when the mass of the body is constant.

Which of the above statements are correct? (a) 1 and 2 only (b) 2 and 3 only (c) 1 and 3 only (d) 1, 2 and 3

Q29. A hydraulic brake works on the principle of (a) Archimedes' principle (b) Bernoulli's principle (c) Newton's third law (d) Pascal's law

Q30. One horsepower is approximately equal to (a) 1,000 W (b) 746 W (c) 500 W (d) 3,600 W

7. Answers and Explanations

7.1 Level 1

Q	Ans	Explanation
1	b	Seven — metre, kilogram, second, ampere, kelvin, mole, candela
2	d	Momentum , being mass times velocity, is a vector. Work, energy and speed are all scalars
3	a	The newton , kg m s^{-2} . The joule is energy, the watt power, the pascal pressure
4	c	Inertia . The first law is often called the law of inertia; the second law defines force quantitatively
5	b	Linear momentum — in every collision, elastic or not. Kinetic energy is conserved only in elastic collisions
6	a	The pascal , N m^{-2}
7	d	90° , because $\cos 90^\circ = 0$
8	c	$\frac{1}{2}mv^2$. (a) is potential energy, (d) is work
9	b	About 11.2 km s^{-1} . Option (a) is the orbital velocity — the two are each other's distractors
10	a	Archimedes' principle

7.2 Level 2

Q	Ans	Explanation
11	c	$\mathbf{v^2 = u^2 + 2as}$ is the time-free equation — the one to reach for whenever time is neither given nor asked
12	d	The bodies move together and the loss of kinetic energy is maximum . Momentum is still conserved, which is what (b) gets wrong
13	a	$\mathbf{e = 1}$ for perfectly elastic; $\mathbf{e = 0}$ for perfectly inelastic
14	b	They act on different bodies and so cannot cancel. If they acted on the same body nothing could ever accelerate
15	c	Displacement . The <i>slope</i> gives acceleration — slope and area are the standard swap
16	a	Both are in free fall with the same acceleration, so there is no contact force. Gravity is present throughout — (b) is the intuitive wrong answer
17	d	Maximum at the poles , minimum at the equator, because of the equatorial bulge and the Earth's rotation
18	b	3,600,000 J — that is $1,000 \text{ W} \times 3,600 \text{ s}$. Option (a) is one watt-hour
19	c	Pascal's law
20	a	Bernoulli's principle — faster flow over the curved upper surface means lower pressure there

7.3 Level 3

Q	Ans	Explanation
21	d	Convert first: $72 \text{ km/h} \times 5/18 = 20 \text{ m s}^{-1}$. Then $a = (20 - 0)/20 = \mathbf{1 \text{ m s}^{-2}}$. Option (a) is what you get by forgetting to convert and dividing 72 by 20
22	b	Momentum before $= 2 \times 5 = 10 \text{ kg m s}^{-1}$. After: $(2 + 3)v = 10$, so $\mathbf{v = 2 \text{ m s}^{-1}}$. Note 15 J of kinetic energy is lost, which is what makes it inelastic
23	c	Work and energy are SCALARS , not vectors. The other three statements are correct
24	a	1 and 2 only . Statement 3 is false twice over — centrifugal force is a pseudo force and it points away from the centre
25	d	A-2 B-1 C-4 D-3 — newton force, pascal pressure, joule energy, watt power
26	b	$v^2 = u^2 - 2gh$ with $v = 0$: $0 = 400 - 20h$, so $h = \mathbf{20 \text{ m}}$
27	c	Zero . The force is vertical and the displacement horizontal, so $\cos 90^\circ = 0$. Effort is not the same as work
28	a	1 and 2 only . Statement 3 is false — the law is general, $F = dp/dt$; it is the <i>form</i> $F = ma$ that assumes constant mass
29	d	Pascal's law — pressure transmitted equally through an enclosed fluid, which is how a small pedal force becomes a large braking force
30	b	About 746 W

THINK LIKE UPSC · THE FOUR SHAPES IN THIS CHAPTER

- The unit-conversion trap.** Q21 offers 3.6 as an option precisely because $72 \div 20 = 3.6$. Convert km/h to m/s **before** you touch an equation
- The scalar-vector reversal.** Q2 and Q23 are the same fact. Work and energy are scalars; momentum and weight are vectors
- The slope-versus-area swap.** Q15. Slope of a velocity–time graph is acceleration; area is displacement
- The intuitive-but-wrong explanation.** Q16 and Q27. Weightlessness is not the absence of gravity, and carrying a load is not work. Where an option matches everyday intuition, check the definition

8. One-Minute Revision

ONE-MINUTE REVISION · CHAPTER F9.1

SEVEN SI BASE UNITS metre · kilogram · second · ampere · kelvin · mole · candela

DERIVED newton = kg m s^{-2} · joule = N m · watt = J s^{-1} · pascal = N m^{-2} · hertz = s^{-1} · dioptre = m^{-1}

SCALARS mass · speed · distance · **work** · **energy** · power · time · temperature · pressure

VECTORS displacement · velocity · acceleration · force · **momentum** · **weight** · impulse

EQUATIONS OF MOTION $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · **$v^2 = u^2 + 2as$ (no time)** · $s = [(u+v)/2]t$

CONVERSION km/h to m/s $\times 5/18$ · 72 km/h = **20 m/s** · 36 km/h = 10 m/s

GRAPHS velocity–time: **slope = acceleration**, **area = displacement**

NEWTON'S LAWS first defines **inertia** · second $F = dp/dt$ in general, $F = ma$ when mass is constant, gives the newton · third: action and reaction act on **DIFFERENT bodies** and **never cancel**

MOMENTUM $p = mv$ · **impulse = $F \times t$ = change in momentum**

COLLISIONS **momentum conserved ALWAYS** · KE conserved **only in elastic** · perfectly inelastic: bodies move together, **maximum KE loss**, **$e = 0$** · elastic **$e = 1$**

CIRCULAR centripetal = mv^2/r **towards** the centre, real · **centrifugal = pseudo force, away from centre**

WORK $W = Fs \cos \theta$ · **zero at 90°** — carrying a bucket on level ground, satellite in circular orbit

ENERGY $KE = \frac{1}{2}mv^2$ · $PE = mgh$ · work–energy theorem: work = change in KE · **1 kWh = 3,600,000 J** · **1 hp about 746 W**

GRAVITATION $F = Gm_1m_2/r^2$ · $g = GM/R^2$ · **g maximum at the POLES** · decreases with altitude and depth · **escape 11.2 km/s** · **orbital 7.9 km/s** · **escape = $\sqrt{2} \times$ orbital**

WEIGHTLESSNESS = FREE FALL, not the absence of gravity

FLUIDS **Pascal** — pressure transmitted equally » hydraulic brakes and press · **Archimedes** — upthrust = weight of fluid displaced » ships, hydrometer · **Bernoulli** — high velocity means low pressure » aerofoil lift, atomiser · **flotation** — a floating body displaces its own weight

9. Five-Minute Wall Sheet

FIVE-MINUTE WALL SHEET · CHAPTER F9.1

× **5/18 FIRST** — km/h to m/s before any equation. 72 km/h = 20 m/s

$v^2 = u^2 + 2as$ — the time-free equation

SLOPE = ACCELERATION · AREA = DISPLACEMENT

WORK AND ENERGY ARE SCALARS

MOMENTUM CONSERVED IN ALL COLLISIONS · KE ONLY IN ELASTIC

PERFECTLY INELASTIC: TOGETHER, MAX LOSS, $e = 0$ · ELASTIC: $e = 1$

ACTION AND REACTION: DIFFERENT BODIES, NEVER CANCEL

$F = dp/dt$ IS GENERAL · $F = ma$ NEEDS CONSTANT MASS

CENTRIFUGAL IS A PSEUDO FORCE, POINTING OUTWARD

WORK = ZERO AT 90° — bucket on level ground, satellite in orbit

1 kWh = 3,600,000 J · 1 hp = 746 W

g MAX AT POLES · ESCAPE 11.2 · ORBITAL 7.9 · ESCAPE = $\sqrt{2}$ × ORBITAL

WEIGHTLESS = FREE FALL, GRAVITY STILL THERE

**MULTIPLIES FORCE » PASCAL · FLOATS » ARCHIMEDES · MOVES THROUGH FLUID »
BERNOULLI**

ONE LINE FOR THE INTERVIEW Momentum is conserved in every collision because conservation follows from Newton's third law, while kinetic energy is conserved only when no energy goes into heat, sound or permanent deformation — which is why a real-world impact is almost always inelastic.

ACTION · NEXT IN MODULE 9

Chapter F9.2 — Physics II: Optics, Heat, Sound, Electricity and Modern Physics. It contains the other two physics items the 2023 paper actually asked: **the power of a combination of lenses** and **a concave mirror with the object at infinity**.

Before you move on, state without looking: the km/h conversion factor; the time-free equation; what is conserved in an inelastic collision; where g is maximum; and which principle explains hydraulic brakes.